HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PB

Probability and Statistics B

Semester 2 2016/17

Duration: Two Hours

ANSWER ALL QUESTIONS

Total Marks: 60

Provided:
Distributions Formula Sheet
Correlation and Regression Formula Sheet
Yellow Actuarial Tables
Blue Statistical Tables

1. (a) The data in Table 1.1, are the scores of 19 candidates in a psychometric test for an actuarial internship.

84	92	63	75	81	97	73	69	38	98
46	58	94	84	78	40	77	82	88	

Table 1.1

Note: $\sum_{i=1}^{19} x_i = 1448$ and $\sum_{i=1}^{19} x_i^2 = 114{,}992$.

(i) Calculate the mean, median, standard deviation and the interquartile range (IQR). Give a brief interpretation of what the IQR measures.

(6 marks)

(ii) Draw a box-and-whisker plot and comment on the skewness of the data. (3 marks)

(iii) Explain why the mean and standard deviation are not necessarily the best statistical representations for this distribution. (1 mark)

(b) The Everlast and Endurance brands of car battery are both labelled as lasting 48 months. In reality, they both have a mean life of 50 months, but Everlast batteries have a standard deviation of 2 months, while Endurance batteries have a standard deviation of 6 months. Which brand is the better choice? Why?

(2 marks)

[Total 12 marks]

2. (a) Suppose that $\underline{X} = (X_1, X_2, ..., X_n)$ is a random sample from a distribution parameterised by unknown θ and let $g(\underline{X})$ denote an estimator for θ.
Give definitions for the *bias* and *mean square error (MSE)* of $g(\underline{X})$.

Express *MSE* $(g(\underline{X}))$ in terms of *Var* $(g(\underline{X}))$ and *Bias* $(g(\underline{X}))$.

(4 marks)

(b) Let $\underline{X} = (X_1, X_2, ..., X_n)$ be a random sample of size n from a distribution with density function

$$f_X(x;\theta) = \frac{xe^{-x/\theta}}{\theta^2} \quad \text{for} \quad x > 0 \text{ (and parameter } \theta > 0).$$

This is a Gamma $(2, \frac{1}{\theta})$ distribution and has mean 2θ and variance $2\theta^2$.

(i) Consider $\frac{\overline{X}}{2}$ as an estimator of θ. Show that $MSE(\frac{\overline{X}}{2}) = \frac{\theta^2}{2n}$ and explain what happens to the *MSE* as n is increased.

(4 marks)

(ii) Now consider $Y = c\overline{X}$ as an estimator of θ where $c > 0$ is a constant.

Show that $MSE(Y) = \frac{2c^2\theta^2}{n} + (2c-1)^2\theta^2$.

(4 marks)

[Total 12 marks]

3. Let $X_1, X_2, \text{K } X_n$ be a random sample from the probability density function

$$f(x) = \begin{cases} (\theta+1)x^{\theta}, & 0 < x < 1, \theta > 0 \\ 0 & \text{, otherwise} \end{cases}$$

(a) Find E(X) for this distribution. (2 marks)

(b) Use the method of moments to find an estimator for θ. (2 marks)

(c) State an advantage and disadvantage of using the method of moments. (2 marks)

(d) Find the maximum-likelihood estimator for θ (4 marks)

[Total 10 marks]

4. A new filtering device is installed in a chemical laboratory. Before its installation, a sample yielded the following information about the percentage of impurity: $\bar{x}_1 = 12.5$, $s_1^2 = 101.17$, and $n_1 = 8$. After installation, a random sample yielded $\bar{x}_2 = 10.2$, $s_2^2 = 94.73$, and $n_2 = 9$. Assume that both populations are normally distributed.

(a) Check the assumption of equal population variances using a suitable 95% confidence interval. Can you conclude that the two variances are equal? Explain.

(4 marks)

(b) (i) Given that $S_p^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$, explain why the function $\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{S_p^2(\frac{1}{n_1} + \frac{1}{n_2})}}$ can be used as a pivotal quantity for μ_1- μ_2.

(2 marks)

(ii) Calculate a 95% confidence interval for the difference in the population means from which the samples come. Comment on the result.

(3 marks)

[Total 9 marks]

5. A medical study is carried out to investigate the average length of time people who catch a cold are affected by the virus. Tests on a sample of 16 patients gave the results in Table 5.1 for the number of days the virus remains in the patient.

9	12	9	7	9	14	16	7	9	10	9	13	11	11	4	9

Table 5.1

Experts on cold viruses claim that the average time that a virus stays in a patient's system is 8 days. However, many patients think that actually they have the virus for longer than this.

(a) Given that for the data in Table 5.1 , $\bar{x} = 9.94$ and $s = 2.91$, carry out a hypothesis test to check whether the virus does seem to last for longer than 8 days. Make sure you state your hypotheses clearly, calculate a p-value and give a clear conclusion. Also check the validity of the test.

(5 marks)

(b) Explain what is meant by Type I and Type II errors in hypothesis testing.

(2 marks)

[Total 7 marks]

6. In a medical experiment concerning 10 patients with a certain type of eye condition, the following measurements were made for blood flow (y) and intraocular pressure (x):

x	10.6	9.8	8.5	9.4	10.1	8.7	11.2	14.2	11.5	13.5
y	10	11	3	7	9	5	8	17	14	13

For these data $\sum x = 107.5$, $\sum x^2 = 1188.29$, $\sum y = 97$, $\sum y^2 = 1103$, $\sum xy = 1107$.

(a) Fit a simple regression model $y_i = \alpha + \beta x_i + \varepsilon_i$, $i = 1, 2, \text{K}, 10$ to the data, where ε_i are the errors.

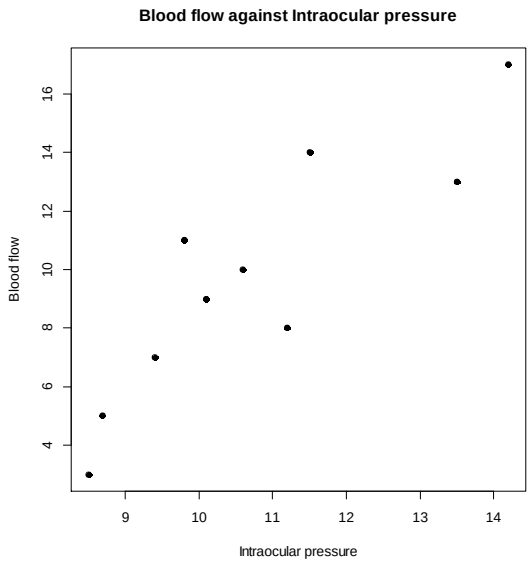
(3 marks)

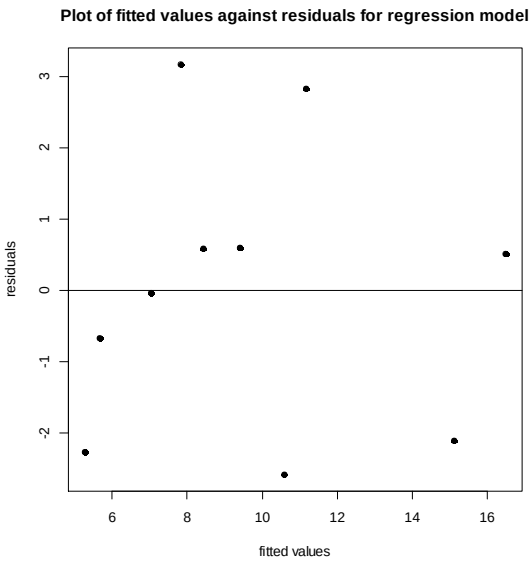
(b) Analyse the plots on the following page.

(i) Comment on any relationship between intraocular pressure and blood flow.

(1 mark)

(ii) Draw conclusions about the model adequacy. (2 marks)





(c) Find the 95% confidence interval for the slope coefficient of the underlying regression line. Using this result, comment on the hypothesis that the true underlying slope coefficient is equal to 1.5? Explain.

(4 marks)

[Total 10 marks]

END OF PAPER